

Andrei Dobrete

Curriculum Vitae

in [linkedin.com/in/Andy3153](https://www.linkedin.com/in/Andy3153)

 github.com/Andy3153

 gitlab.com/Andy3153



*Self-taught Systems Engineer and Robotics Engineering Graduate focused on network/server management, Linux, NixOS, Docker and shell/Python automation.
Hands-on experience with PLC factory automation and web development.*

Work Experience

07/2025 – 08/2025

Automation/Electrical Engineer – internship

IMSAT S.A. | imsat.ro

- Implemented electrical and control-system modifications at the Ford Otosan plant in Craiova, commissioning and troubleshooting conveyor systems used for door-accessory assembly.
- Programmed Allen-Bradley PLCs using software like RSLogix 5000 and developed operator interfaces (FactoryTalk View Studio for PanelView Plus 1000) also used Connected Components Workbench for supporting controllers.
- Created and revised 3D CAD models (Siemens NX, CATIA V5) to validate component fit and support mechanical-controls integration.

06/2024 – 08/2024

Web Developer – internship

EVASTAFF | evastaff.ro

- Implemented and maintained WordPress and Shopify e-commerce platforms on dedicated hosting using cPanel, including server configuration and deployment.
- Designed and developed complete online stores: storefronts, product listings, shopping cart and checkout flows.
- Automated sales and order-fulfillment processes to streamline the selling lifecycle.
- Performed front- and back-end customizations (PHP, CSS, SQL) and developed custom themes/plugins.
- Integrated third-party services and payment gateways via APIs to connect inventory, shipping and CRM systems.

Education and Training

03/2025 – Present

Cisco Certified Network Associate (CCNA)

Hackademy | hackademy7.wordpress.com

- Completed Semesters 1 & 2, covering Network Fundamentals, Switching, Routing, and Wireless Essentials.

10/2022 – 06/2026

Bachelor of Engineering (B.Eng.) in Robotics

Universitatea Națională de Știință și Tehnologie Politehnica București –
Faculty of Industrial Engineering and Robotics | fiir.upb.ro

- **Thesis:** "Robotic palletizing cell with two inputs and two outputs integrating an articulated arm industrial robot equipped with a multifunctional effector"

Publications

05/2023

Phone Controlled Car | github.com/Andy3153/phone-controlled-car

Submissions:

- Student Scientific Communications Session of the Faculty of Industrial Engineering and Robotics, 2023 (industrialengineeringjournal.eu/pdf/vol7_2023_4/Art9.pdf)
- Journal of Industrial Engineering and Robotics, Volume 7, Number 4, 2023 (industrialengineeringjournal.eu/pdf/vol7_2023_4/vol7_2023_4.pdf, page 54)

Digital Skills

Linux/Unix	○ System Administration	○ Docker/Podman	○ NixOS
	○ Shell Scripting	○ Systemd	
	○ Command Line	○ QEMU/KVM	
Programming	○ C	○ Nix Language	○ Arduino
	○ C++	○ Python	○ Java
	○ POSIX Shell (bash/sh)	○ Lua	○ Ladder logic
CAD/CAM	○ Siemens NX	○ AutoCAD	○ FreeCAD
	○ CATIA V5	○ Blender	
Office	○ L ^A T _E X	○ Microsoft Office	○ LibreOffice

Projects

10/2019 – 10/2022	Lead Programmer, BroBots (First Tech Challenge RO142 19088) ○ Primary developer for the robot control codebase using Java, responsible for implementation, debugging and version control. ○ Heavily contributed to mechanical design, assembly and prototyping. ○ Contributed to iterative testing and performance tuning for competition readiness. ○ Mentored newer team members and facilitated cross-functional collaboration between software and mechanical subteams.
Present	Systems Engineering & Self-Hosted Infrastructure ○ Design and manage NixOS configurations for multi-machine environments (my personal computer and my home server), ensuring consistency and declarative system management. ○ Author and maintain custom Nix derivations and reproducible development environments. ○ Build and operate Docker Compose stacks and container images for self-hosted server deployments. ○ Implement shell and Python automation scripts to streamline daily system operations and maintenance tasks. ○ Curate and version-control application configurations (dotfiles) and L^AT_EX templates for portability and reproducibility. ○ Manage upstream collaboration by maintaining forks, preparing patches, and submitting pull requests.

Volunteering

03/2024 – Present	Volunteers Polytechnic Bucharest – Member unstopb.ro
10/2022 – 06/2023	The Student Organization of the Faculty of Industrial Engineering and Robotics – Member osfiir.ro
07/2019 – 12/2022	National Red Cross Society Râmnicu Vâlcea – Member crucearosievalcea.ro

Language Skills

Romanian	Native
English	B2 (Professional Working Proficiency)

Driving License

08/2021	Full EU – Categories B , B1 , AM
---------	---